

Nibrexo OS AI

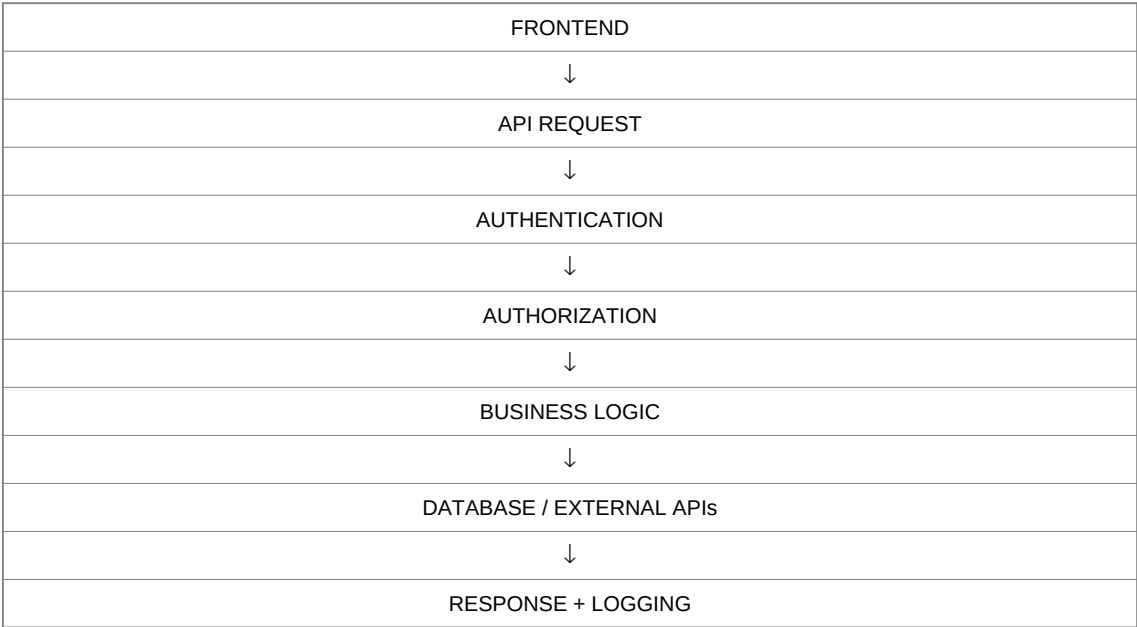
06 — Backend Architecture & API Design v1.0

Production backend blueprint for Nibrexo OS AI using Supabase, secure APIs and AI agent orchestration.

1. Backend Architecture Philosophy

- Backend is the security and business logic layer between frontend and external services.
- Frontend never directly handles privileged operations.
- All external integrations use controlled service modules.
- Every important action creates logs for tracking.
- Architecture must support future scaling.

2. High Level Backend Flow

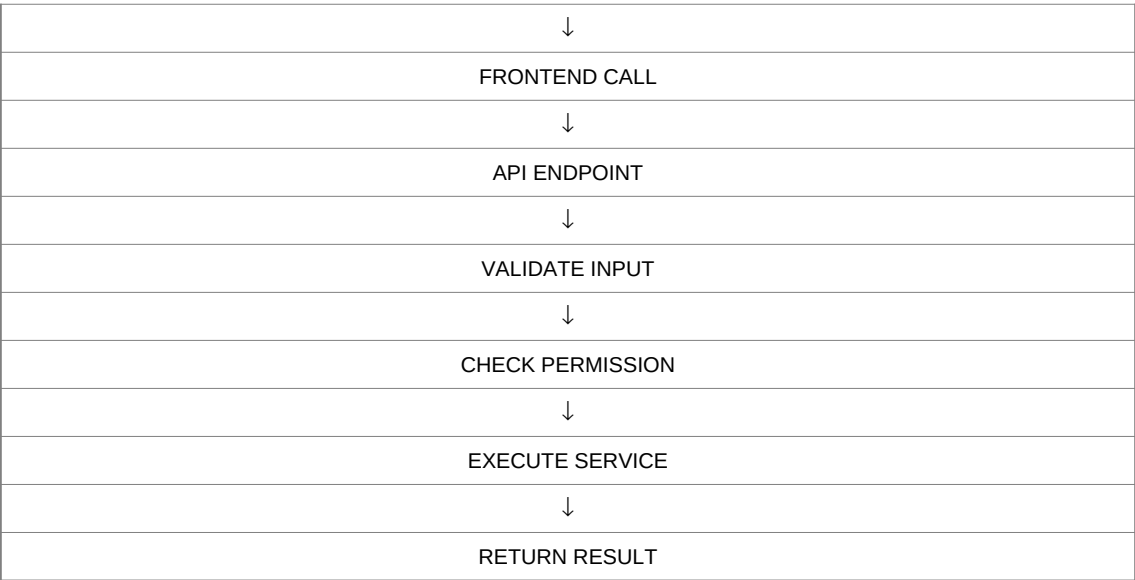


3. Backend Core Layers

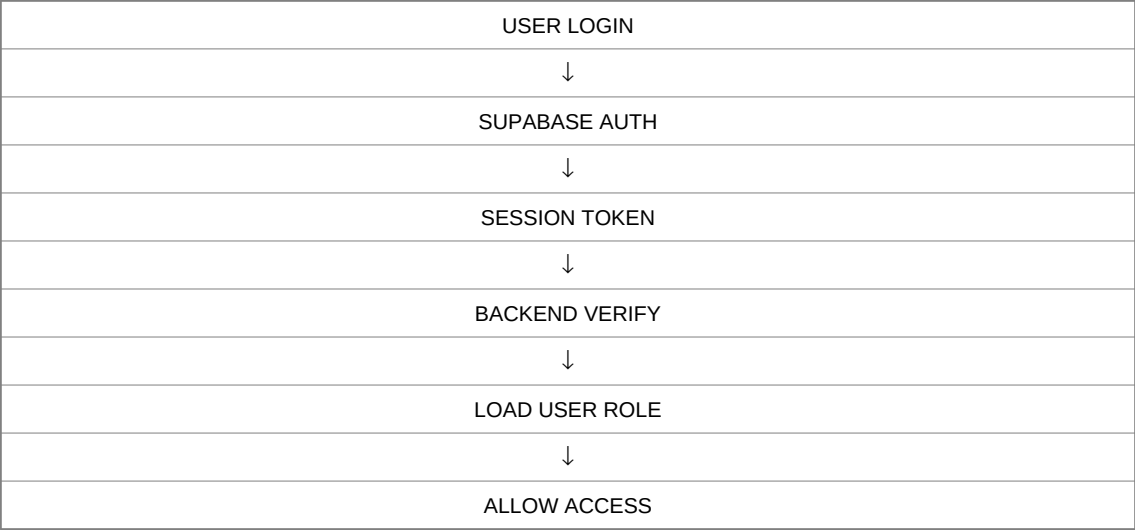
Layer	Responsibility
API Layer	Receives requests and returns responses
Auth Layer	Identity and permissions
Service Layer	Business logic
Integration Layer	Social/email/platform APIs
Database Layer	Supabase queries and transactions
AI Layer	Agent tools and AI operations
Logging Layer	Audit and debugging

4. API Request Lifecycle





5. Authentication Architecture



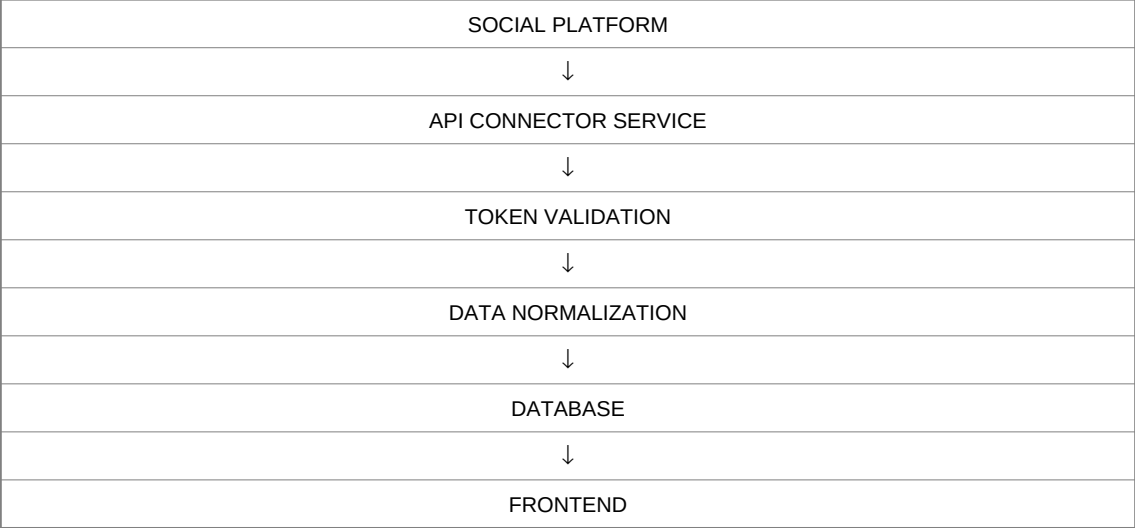
- Authentication handled through Supabase Auth.
- Sessions are verified before protected operations.
- Organization and role permissions are checked server-side.
- Sensitive operations require additional authorization.

6. API Module Structure

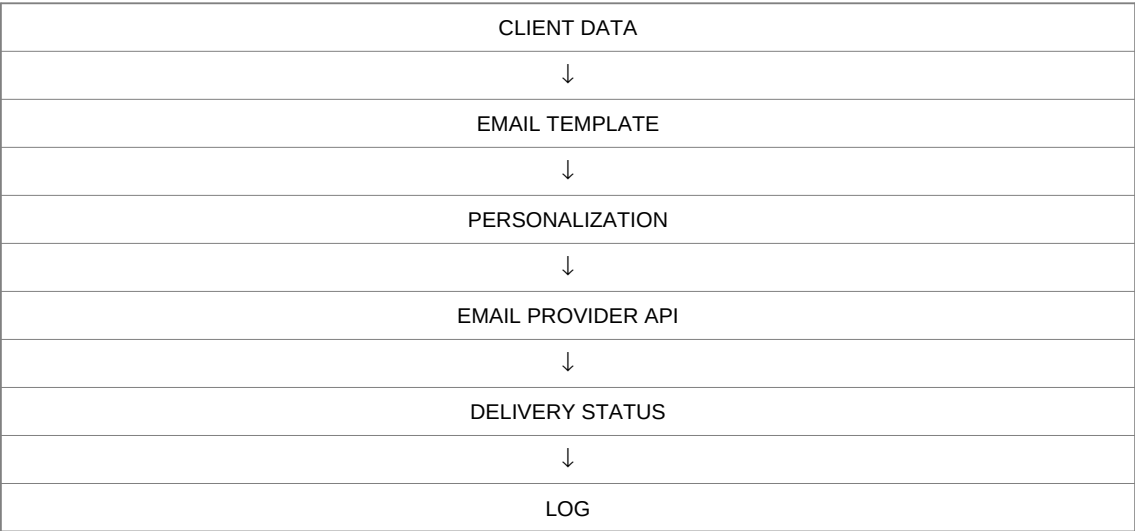
Module	Example Responsibility
Auth API	Login, session, user profile
Organization API	Workspace management
Social API	Accounts, posts, sync
Content API	Media, drafts, publishing
Inbox API	Messages/comments
CRM API	Clients and activities
Email API	Templates and automation

Module	Example Responsibility
AI API	Agent requests and tools
Settings API	Configuration

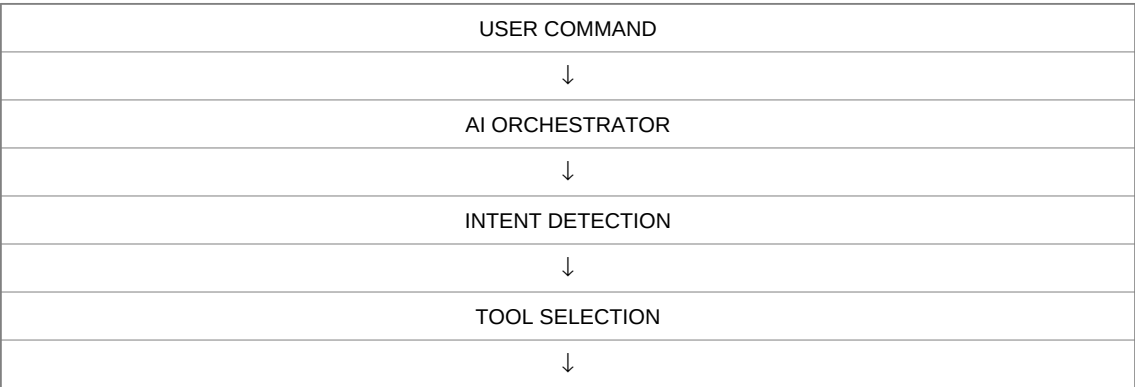
7. Social Integration Architecture

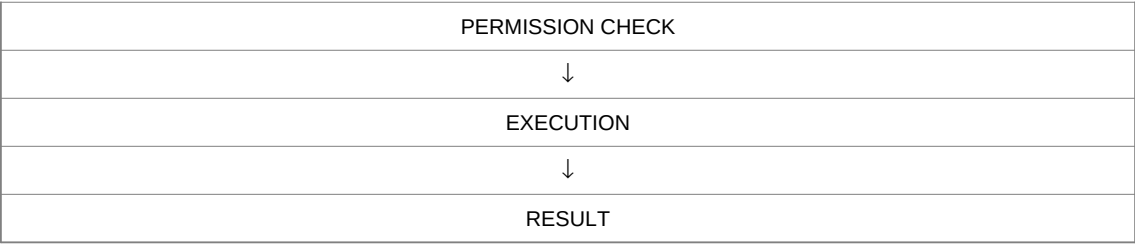


8. Email Service Architecture



9. AI Agent Backend Architecture





10. Background Jobs & Automation

- Scheduled publishing jobs.
- Email sequence execution.
- Social data synchronization.
- Notification processing.
- AI task processing.
- Retry failed external operations.

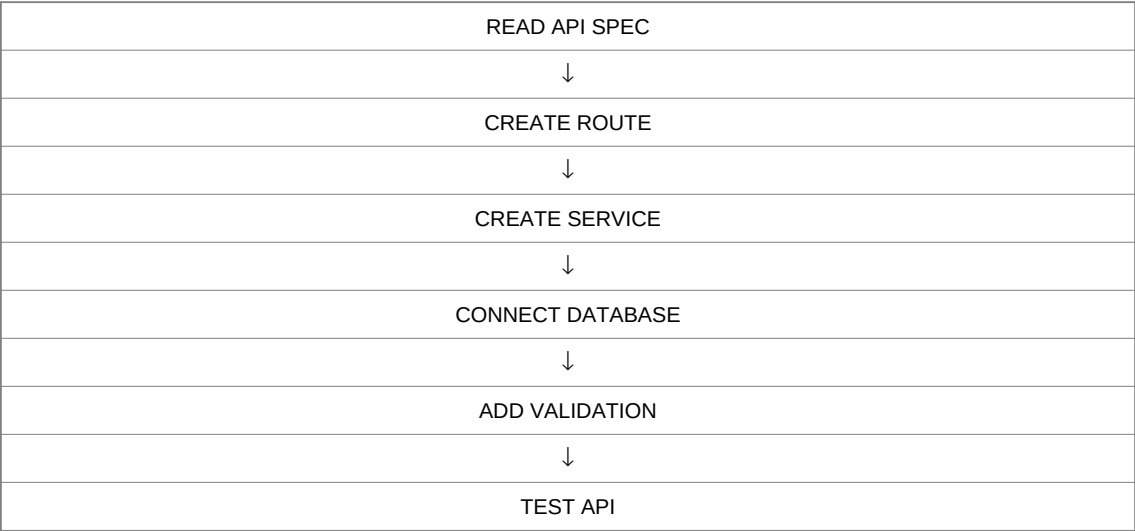
11. Error Handling Strategy

- Every error has a readable message.
- External API failures are logged.
- Retries are controlled to avoid duplication.
- Users see actionable error states.
- Sensitive technical details are hidden from users.

12. Security Architecture

- Environment variables for secrets.
- Server-side API keys only.
- Input validation on every endpoint.
- Rate limiting for abuse prevention.
- Audit logs for important operations.

13. LLM Arena Development Flow



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DOCUMENT CHANGE

Document Version 1.0 | Nibrexo OS AI | Backend Architecture & API Design